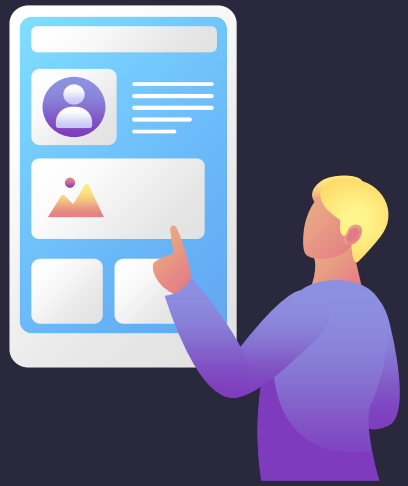
Factor & Component Analysis

Week 3 – Data Mining I
TSD -FTMM

/01

/BASIC CONCEPTS



/Why Factor or Component Analysis?

- We study phenomena that can not be directly observed
  - ego, personality, intelligence in psychology
  - Underlying factors that govern the observed data
- We want to identify and operate with underlying latent factors rather than the observed data
  - E.g. topics in news articles
  - Transcription factors in genomics
- We want to discover and exploit hidden relationships
  - “beautiful car” and “gorgeous automobile” are closely related
  - So are “driver” and “automobile”
  - But does your search engine know this?
  - Reduces noise and error in results

/Why Factor or Component Analysis?

- We have too many observations and dimensions
  - To reason about or obtain insights from
  - To visualize
  - Too much noise in the data
  - Need to “reduce” them to a smaller set of factors
  - Better representation of data without losing much information
  - Can build more effective data analyses on the reduced-dimensional space: classification, clustering, pattern recognition

- Combinations of observed variables may be more effective bases for insights, even if physical meaning is obscure

Factor or Component Analysis

- Discover a new set of factors/dimensions/axes against which to represent, describe or evaluate the data
  - For more effective reasoning, insights, or better visualization
  - Reduce noise in the data
  - Typically a smaller set of factors: dimension reduction
  - Better representation of data without losing much information
  - Can build more effective data analyses on the reduced-dimensional space: classification, clustering, pattern recognition

- Factors are combinations of observed variables
  - May be more effective bases for insights, even if physical meaning is obscure
  - Observed data are described in terms of these factors rather than in terms of original variables/dimensions

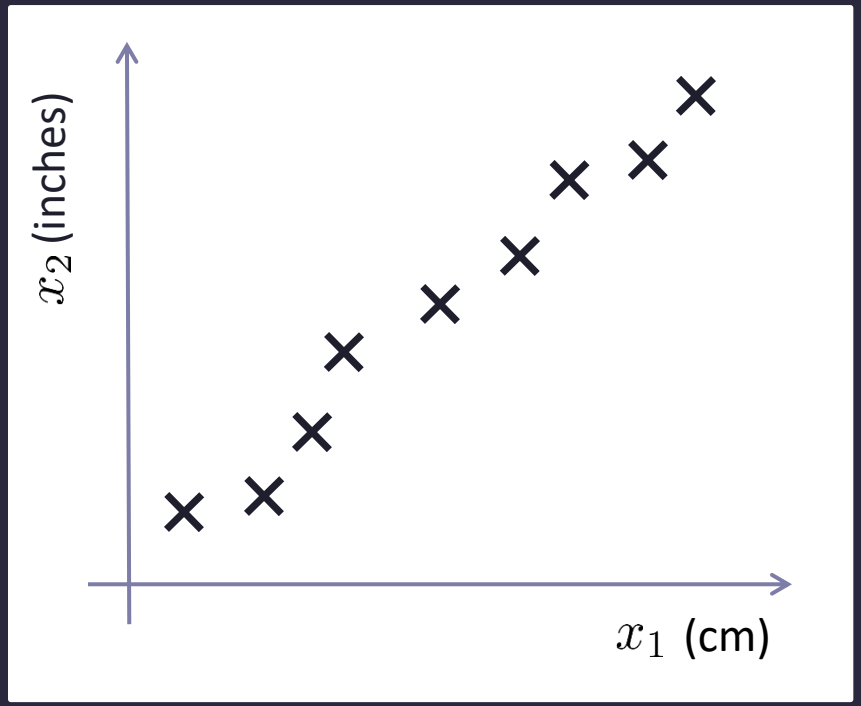
Basic Concept

- Areas of variance in data are where items can be best discriminated and key underlying phenomena observed
  - Areas of greatest “signal” in the data
- If two items or dimensions are highly correlated or dependent
  - They are likely to represent highly related phenomena
  - If they tell us about the same underlying variance in the data, combining them to form a single measure is reasonable
    - Parsimony
    - Reduction in Error
- So we want to combine related variables, and focus on uncorrelated or independent ones, especially those along which the observations have high variance
- We want a smaller set of variables that explain most of the variance in the original data, in more compact and insightful form

Basic Concept

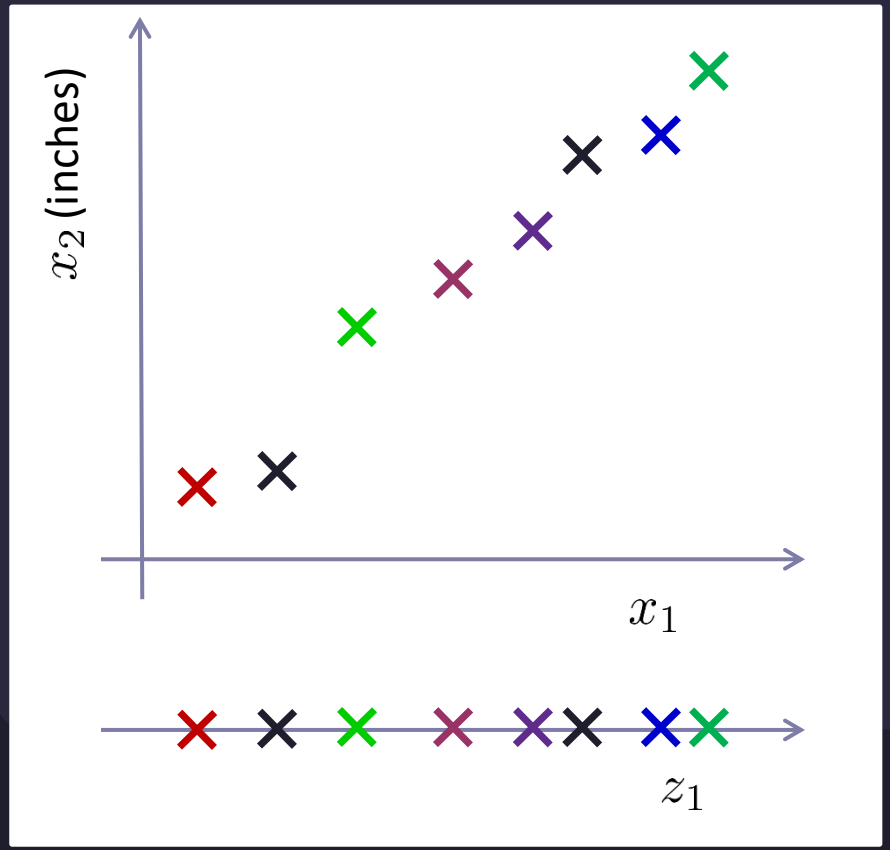
- What if the dependences and correlations are not so strong or direct?
- And suppose you have 3 variables, or 4, or 5, or 10000?
- Look for the phenomena underlying the observed covariance/co-dependence in a set of variables
  - Once again, phenomena that are uncorrelated or independent, and especially those along which the data show high variance
- These phenomena are called “factors” or “principal components” or “independent components,” depending on the methods used
  - Factor analysis: based on variance/covariance/correlation
  - Independent Component Analysis: based on independence

Data Compression



Reduce data from
2D to 1D

Data Compression

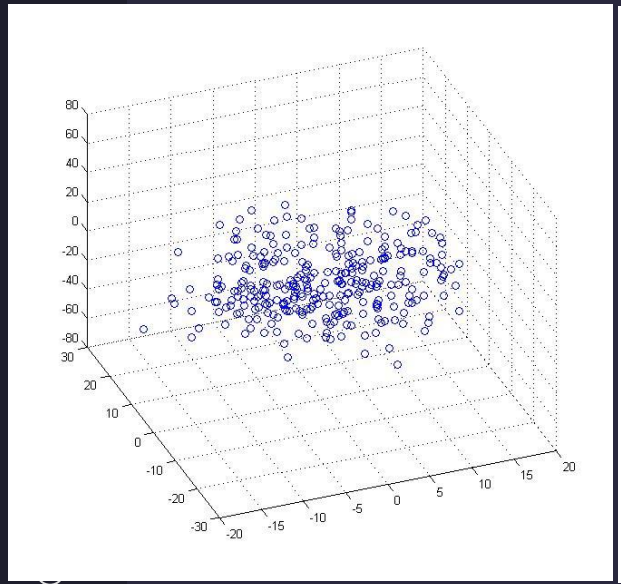


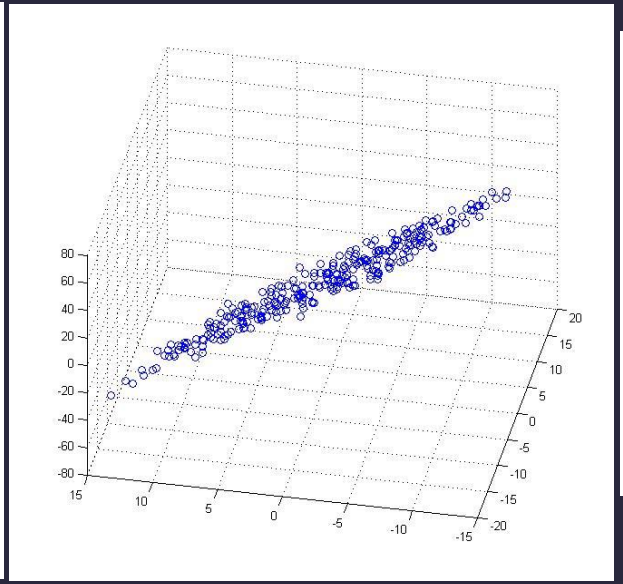
Reduce data from 2D to 1D

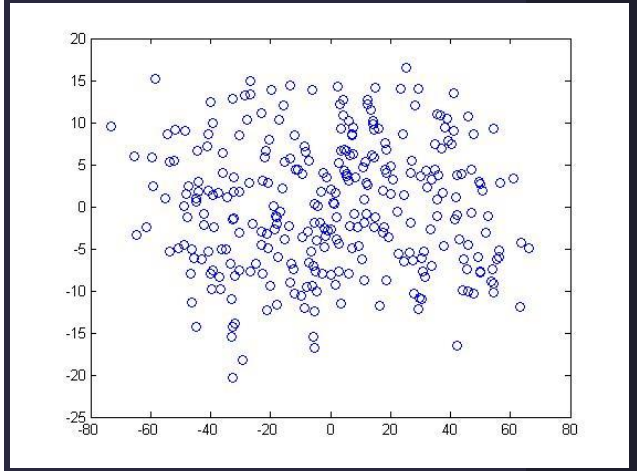
$$
\begin{aligned}
x^{(1)} &\rightarrow z^{(1)} \\
x^{(2)} &\rightarrow z^{(2)} \\
&\vdots \\
x^{(m)} &\rightarrow z^{(m)}
\end{aligned}
$$

Data Compression

Reduce data from 3D to 2D







Andrew Ng

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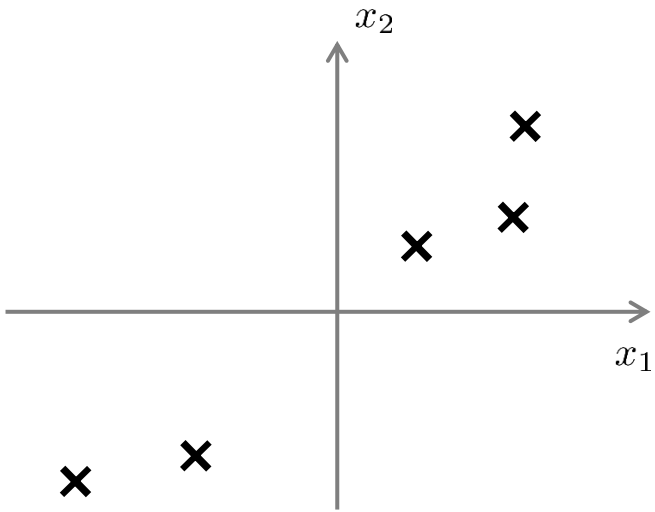
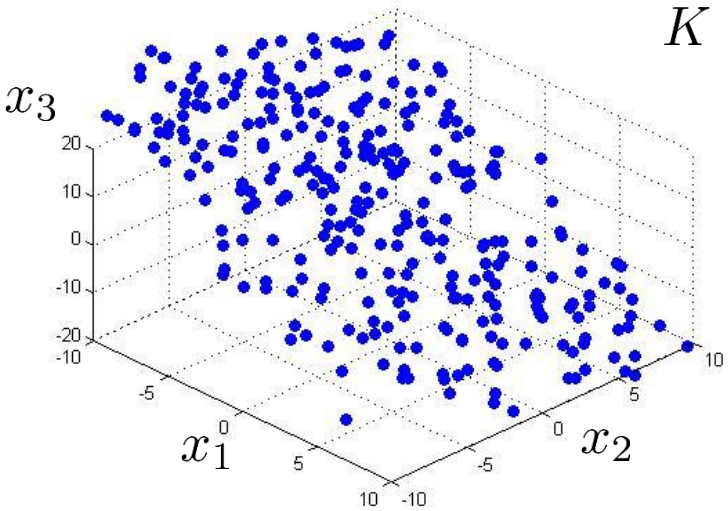
Principal Component Analysis (PCA) problem formulation

$$3D \rightarrow 2D$$
$$K = 2$$





Reduce from 2-dimension to 1-dimension: Find a direction (a vector $u^{(1)} \in \mathbb{R}^n$) onto which to project the data so as to minimize the projection error.

Reduce from n-dimension to k-dimension: Find k vectors $u^{(1)}, u^{(2)}, \ldots, u^{(k)}$ onto which to project the data, so as to minimize the projection error.

Principal Component Analysis

Goal: Find r-dim projection that best preserves variance

1. Compute mean vector μ and covariance matrix Σ of original points
2. Compute eigenvectors and eigenvalues of Σ
3. Select top r eigenvectors
4. Project points onto subspace spanned by them:

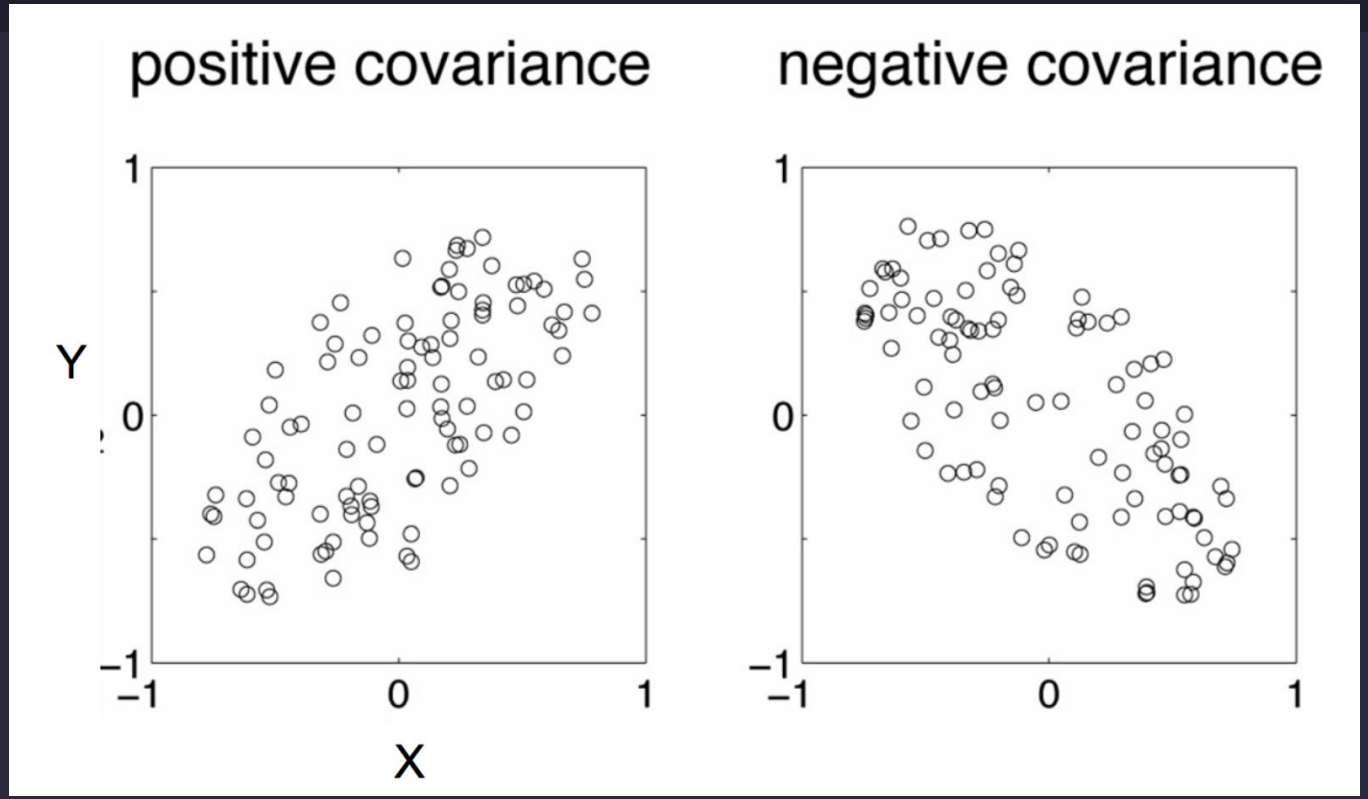
$$y = A(x - \mu)$$

where y is the new point, x is the old one, and the rows of A are the eigenvectors

Covariance

- Variance and Covariance:
  Measure of the “spread” of a set of points around their center of mass(mean)
- Variance:
  Measure of the deviation from the mean for points in one dimension
- Covariance:
  Measure of how much each of the dimensions vary from the mean with respect to each other

- **Covariance is measured between two dimensions**
- **Covariance sees if there is a relation between two dimensions**
- **Covariance between one dimension is the variance**



Positive: Both dimensions increase or decrease together

Negative: While one increase the other decrease

Covariance

Used to find relationships between dimensions in high dimensional data sets

$$q_{jk} = \frac{1}{N}\sum_{i=1}^{N}(X_{ij} - E(X_j))(X_{ik} - E(X_k))$$

The Sample mean

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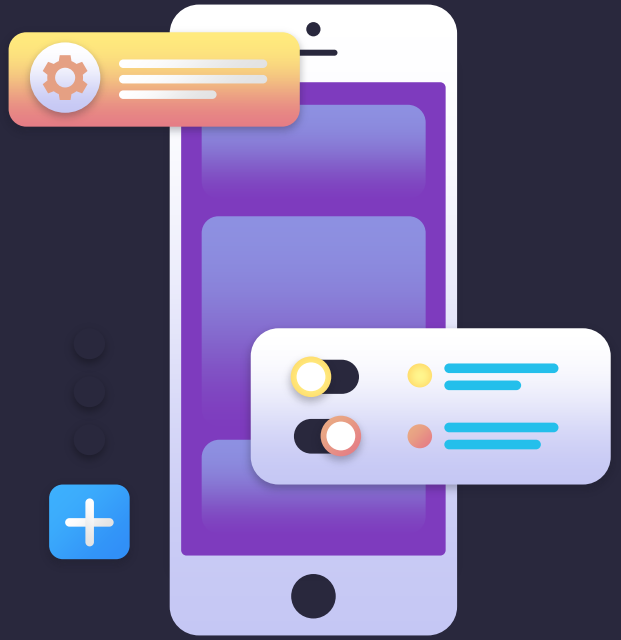
/PRINCIPAL COMPONENT ANALYSIS



Principal Component Analysis

Input : $\mathbf{x} \in \mathbb{R}^D$: $\mathcal{D} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$

Set of basis vectors: $\mathbf{u}_1, \ldots, \mathbf{u}_K$

Summarize a *D* dimensional vector *X* with *K* dimensional feature vector *h(x)*

$$h(\mathbf{x}) = \begin{bmatrix} \mathbf{u}_1 \cdot \mathbf{x} \\ \mathbf{u}_2 \cdot \mathbf{x} \\ \cdots \\ \mathbf{u}_K \cdot \mathbf{x} \end{bmatrix}$$

Principal Component Analysis

$$\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_K]$$

Basis vectors are orthonormal

$$\mathbf{u}_i^T \mathbf{u}_j = 0$$

$$||\mathbf{u}_j|| = 1$$

New data representation *h(x)*

$$z_j = \mathbf{u}_j \cdot \mathbf{x}$$

$$h(\mathbf{x}) = [z_1, \ldots, z_K]^T$$

Principal Component Analysis

$$\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_K]$$

New data representation
h(x)

$$h(\mathbf{x}) = \mathbf{U}^T \mathbf{x}$$

$$h(\mathbf{x}) = \mathbf{U}^T (\mathbf{x} - \mu_0)$$

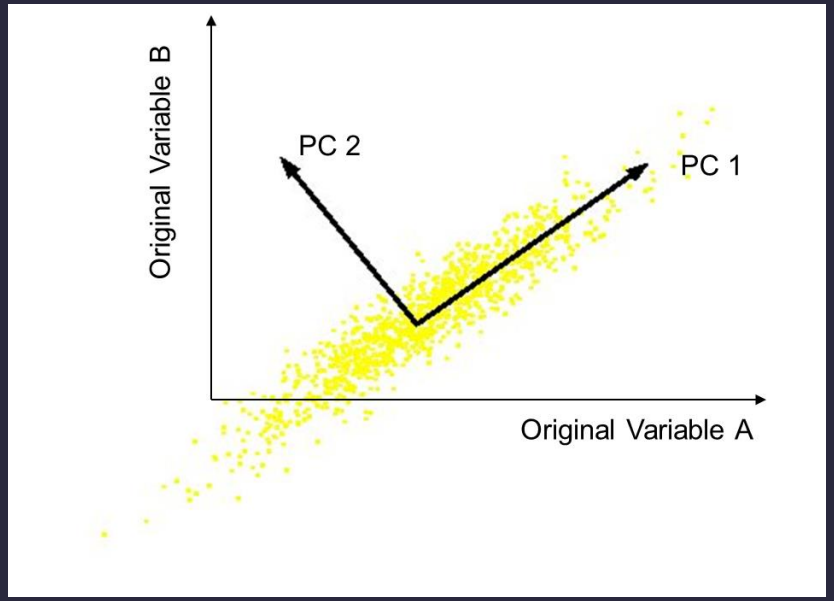
Empirical mean of the data

$$\mu_0 = \frac{1}{N} \sum_{i=1}^{N} \mathbf{x}_i$$

Principal Components Analysis (PCA)

- Most common form of factor analysis
- The new variables/dimensions
  - Are linear combinations of the original ones
  - Are uncorrelated with one another
  - Orthogonal in original dimension space
  - Capture as much of the original variance in the data as possible
  - Are called Principal Components

What are the new axes?



- Orthogonal directions of greatest variance in data
- Projections along PC1 discriminate the data most along any one axis

Principal Components Analysis (PCA)

- First principal component is the direction of greatest variability (covariance) in the data
- Second is the next orthogonal (uncorrelated) direction of greatest variability
  - So first remove all the variability along the first component, and then find the next direction of greatest variability
- And so on …

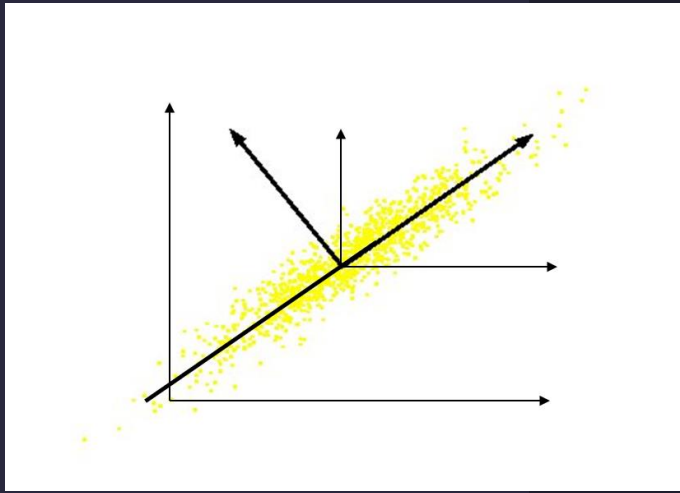
Principal Components Analysis (PCA)

- Principle
  - Linear projection method to reduce the number of parameters
  - Transfer a set of correlated variables into a new set of uncorrelated variables
  - Map the data into a space of lower dimensionality
  - Form of unsupervised learning

- Properties
  - It can be viewed as a rotation of the existing axes to new positions in the space defined by original variables
  - New axes are orthogonal and represent the directions with maximum variability

Computing the Components

- Data points are vectors in a multidimensional space
- Projection of vector x onto an axis (dimension) u is u.x
- Direction of greatest variability is that in which the average square of the projection is greatest
  - I.e. u such that E((u.x)2) over all x is maximized
  - (we subtract the mean along each dimension, and center the original axis system at the centroid of all data points, for simplicity)
  - This direction of u is the direction of the first Principal Component



Computing the Components

- $E((\mathbf{u}.\mathbf{x})^2) = E((\mathbf{u}.\mathbf{x})(\mathbf{u}.\mathbf{x})^T) = E(\mathbf{u}.\mathbf{x}.\mathbf{x}^T.\mathbf{u}^T)$
- The matrix $\mathbf{C} = \mathbf{x}.\mathbf{x}^T$ contains the correlations (similarities) of the original axes based on how the data values project onto them
- So we are looking for w that maximizes $\mathbf{u}\mathbf{C}\mathbf{u}^T$, subject to **u** being unit-length
- It is maximized when w is the principal eigenvector of the matrix **C**, in which case
  - $\mathbf{u}\mathbf{C}\mathbf{u}^T = \mathbf{u}\lambda\mathbf{u}^T = \lambda$ if **u** is unit-length, where λ is the principal eigenvalue of the correlation matrix C
  - The eigenvalue denotes the amount of variability captured along that dimension

Why the Eigenvectors?

Maximise $\mathbf{u}^T\mathbf{x}\mathbf{x}^T\mathbf{u}$ s.t $\mathbf{u}^T\mathbf{u} = 1$

Construct Langrangian $\mathbf{u}^T\mathbf{x}\mathbf{x}^T\mathbf{u} - \lambda\mathbf{u}^T\mathbf{u}$

Vector of partial derivatives set to zero

$$\mathbf{x}\mathbf{x}^T\mathbf{u} - \lambda\mathbf{u} = (\mathbf{x}\mathbf{x}^T - \lambda\mathbf{I})\,\mathbf{u} = 0$$

As $\mathbf{u} \neq \mathbf{0}$ then $\mathbf{u}$ must be an eigenvector of $\mathbf{x}\mathbf{x}^T$ with eigenvalue λ

Singular Value Decomposition

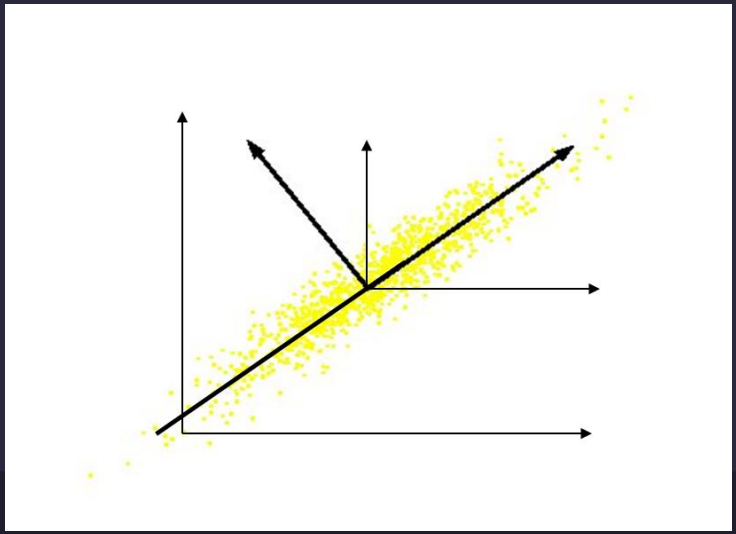
The first root is called the prinicipal eigenvalue which has an associated orthonormal ($\mathbf{u}^T\mathbf{u} = 1$) *eigenvector* $\mathbf{u}$

Subsequent roots are ordered such that $\lambda_1 > \lambda_2 > \ldots > \lambda_M$ with rank($\mathbf{D}$) non-zero values

Eigenvectors form an orthonormal basis i.e. $\mathbf{u}_i^T\mathbf{u}_j = \delta_{ij}$
The eigenvalue decomposition of $\mathbf{x}\mathbf{x}^T = \mathbf{U}\boldsymbol{\Sigma}\mathbf{U}^T$
where $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_M]$ and $\boldsymbol{\Sigma} = \mathrm{diag}[\lambda_1, \lambda_2, \ldots, \lambda_M]$
Similarly the eigenvalue decomposition of $\mathbf{x}^T\mathbf{x} = \mathbf{V}\boldsymbol{\Sigma}\mathbf{V}^T$
The SVD is closely related to the above $\mathbf{x}=\mathbf{U}\,\boldsymbol{\Sigma}^{1/2}\,\mathbf{V}^T$
The left eigenvectors $\mathbf{U}$, right eigenvectors $\mathbf{V}$,
singular values = square root of eigenvalues.

Computing the Components

- Similarly for the next axis, etc.
- So, the new axes are the eigenvectors of the matrix of correlations of the original variables, which captures the similarities of the original variables based on how data samples project to them



- Geometrically: centering followed by rotation
  - Linear transformation

PCs, Variance and Least-Squares

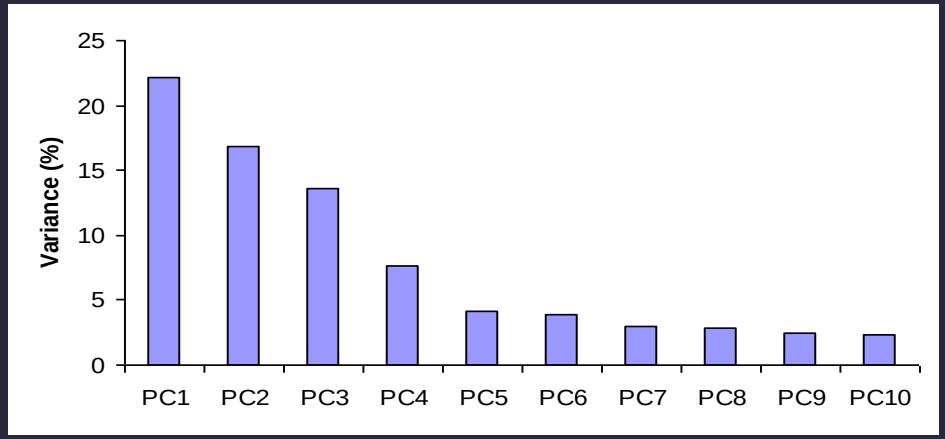
- The first PC retains the greatest amount of variation in the sample
- The kth PC retains the kth greatest fraction of the variation in the sample
- The kth largest eigenvalue of the correlation matrix C is the variance in the sample along the kth PC
- The least-squares view: PCs are a series of linear least squares fits to a sample, each orthogonal to all previous ones

How Many PCs?

- For n original dimensions, correlation matrix is nxn, and has up to n eigenvectors. So n PCs.
- Where does dimensionality reduction come from?

Dimensionality Reduction

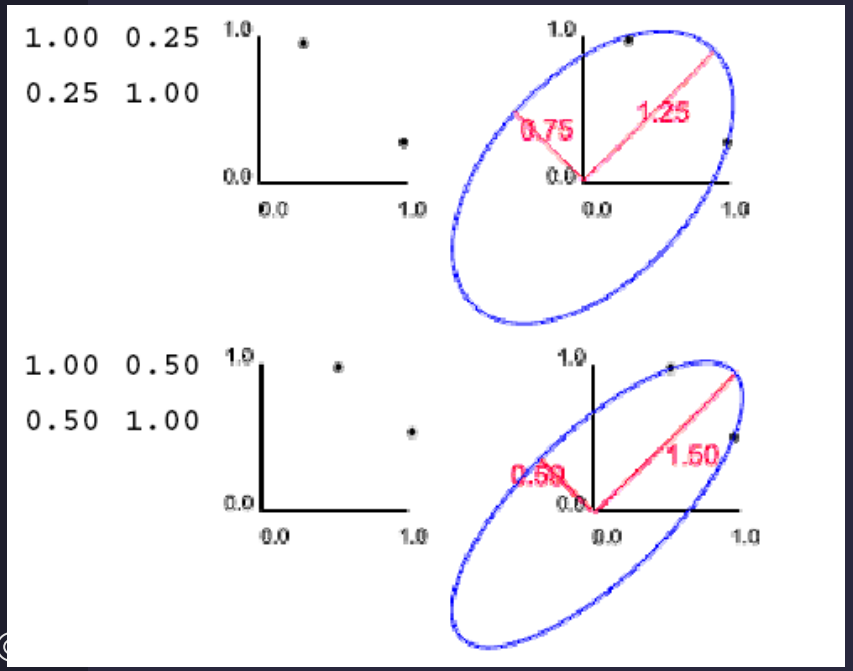
Can ignore the components of lesser significance.

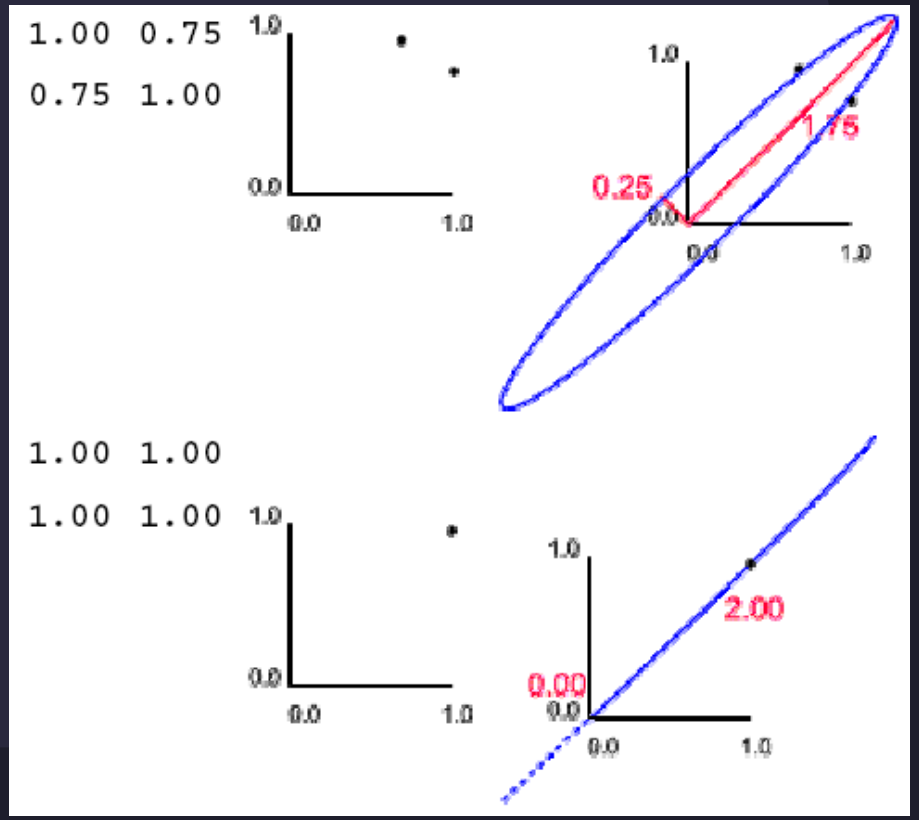


You do lose some information, but if the eigenvalues are small, you don’t lose much

- n dimensions in original data
- calculate n eigenvectors and eigenvalues
- choose only the first p eigenvectors, based on their eigenvalues
- final data set has only p dimensions

Eigenvectors of a Correlation Matrix





PCA / FA

- Principal Components Analysis
  - Extracts all the factor underlying a set of variables
  - The number of factors = the number of variables
  - Completely explains the variance in each variable
  - Parsimony?!?!?!

- Factor Analysis
  - Also known as Principal Axis Analysis
  - Analyses only the shared variance
    - NO ERROR!

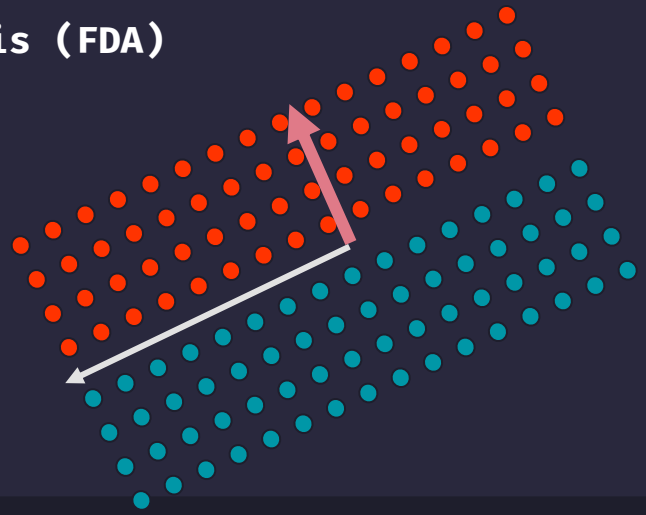
Factor Analysis (e.g. PCA) is not the most sophisticated dimensionality reduction technique

- Dimensionality reduction is a useful technique for any classification/regression problem
  - Text retrieval can be seen as a classification problem
- There are many other dimensionality reduction techniques
  - Neural nets, support vector machines etc.

Limitations of PCA

Are the maximal variance dimensions the relevant dimensions for preservation?

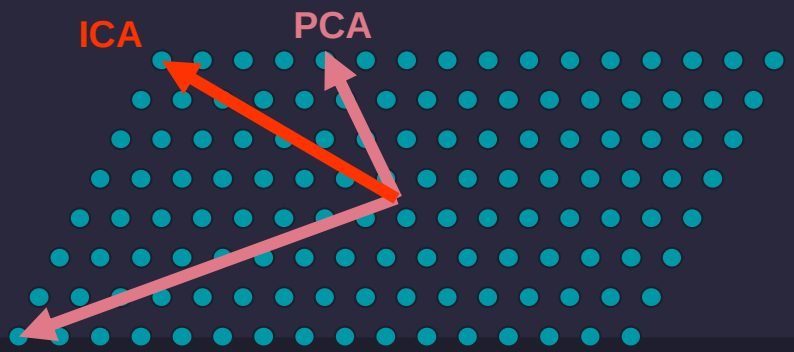
- Relevant Component Analysis (RCA)
- Fisher Discriminant analysis (FDA)



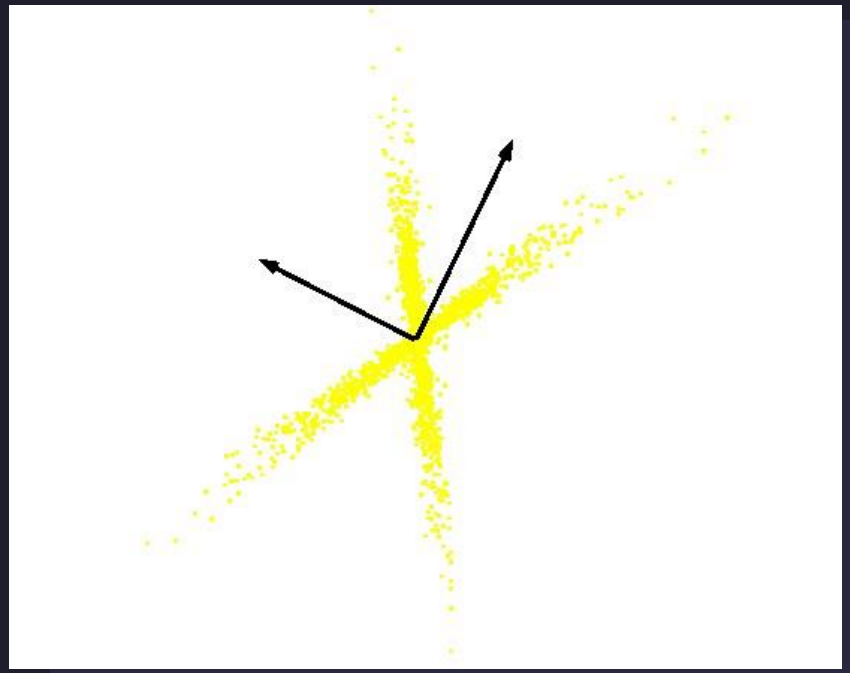
Limitations of PCA

Should the goal be finding independent rather than pair-wise uncorrelated dimensions

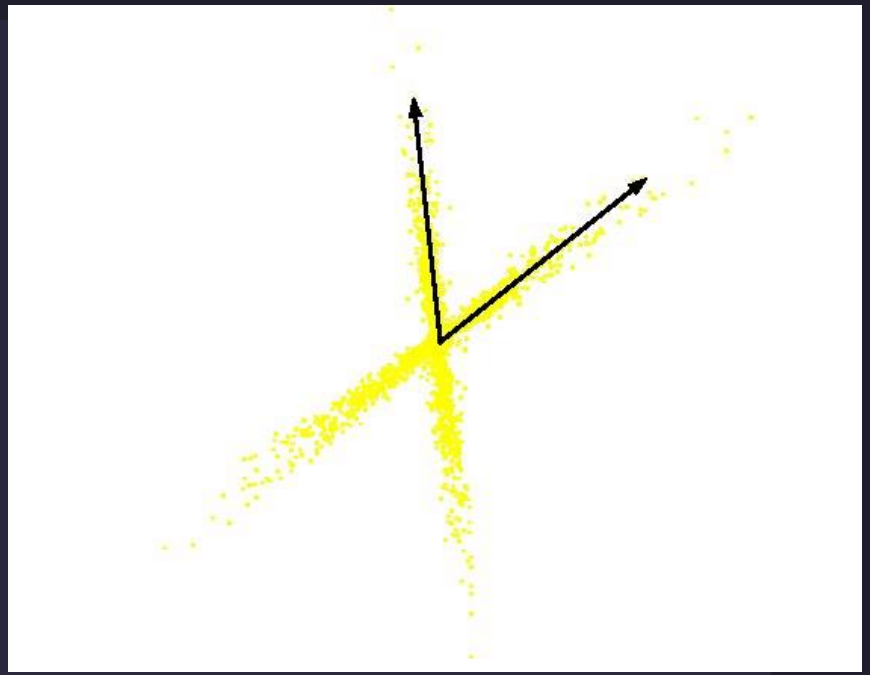
- Independent Component Analysis (ICA)



PCA vs ICA



PCA
(orthogonal coordinate)

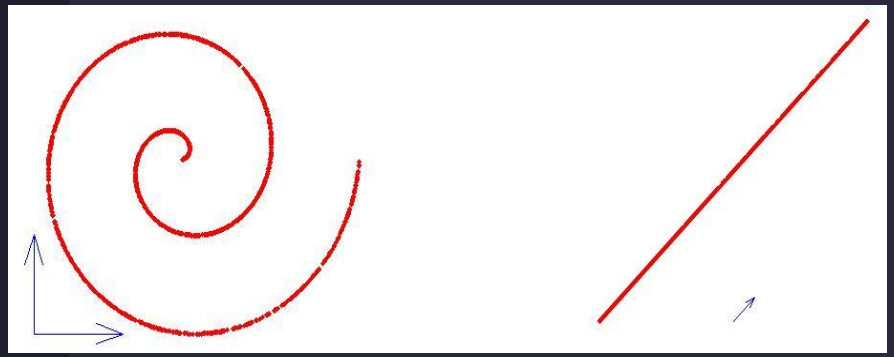


ICA
(non-orthogonal coordinate)

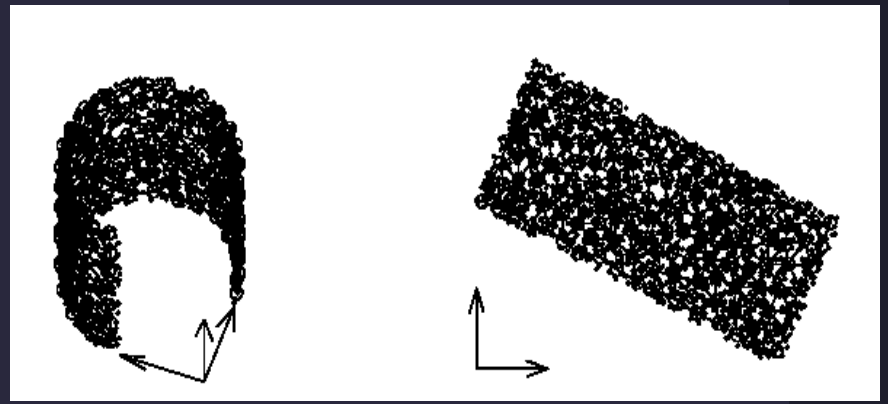
Limitations of PCA

- The reduction of dimensions for complex distributions may need non linear processing
- Curvilenear Component Analysis (CCA)
  - Non linear extension of PCA
  - Preserves the proximity between the points in the input space i.e. local topology of the distribution
  - Enables to unfold some varieties in the input data
  - Keep the local topology

Example of data representation using CCA



Non linear projection of a spiral



Non linear projection of a horseshoe

PCA applications -Eigenfaces

- Eigenfaces are
  the eigenvectors of
  the covariance matrix of
  the probability distribution of
  the vector space of
  } human faces

- Eigenfaces are the ‘standardized face ingredients’ derived from the statistical analysis of many pictures of human faces
- A human face may be considered to be a combination of these standard faces

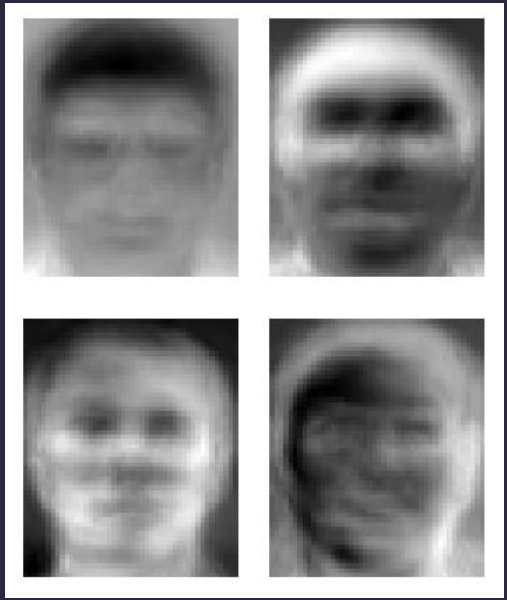
PCA applications -Eigenfaces

To generate a set of eigenfaces:

1. Large set of digitized images of human faces is taken under the same lighting conditions.
2. The images are normalized to line up the eyes and mouths.
3. The eigenvectors of the covariance matrix of the statistical distribution of face image vectors are then extracted.
4. These eigenvectors are called eigenfaces.

PCA applications -Eigenfaces

the principal eigenface looks like a bland androgynous average human face



http://en.wikipedia.org/wiki/Image:Eigenfaces.png

Eigenfaces – Face Recognition

- When properly weighted, eigenfaces can be summed together to create an approximate gray-scale rendering of a human face.
- Remarkably few eigenvector terms are needed to give a fair likeness of most people's faces
- Hence eigenfaces provide a means of applying data compression to faces for identification purposes.
- Similarly, Expert Object Recognition in Video

Eigenfaces

Experiment and Results:
Data used here are from the ORL database of faces. Facial images of 16 persons each with 10 views are used.

- Training set contains 16×7 images.
- Test set contains 16×3 images.

First three eigenfaces :

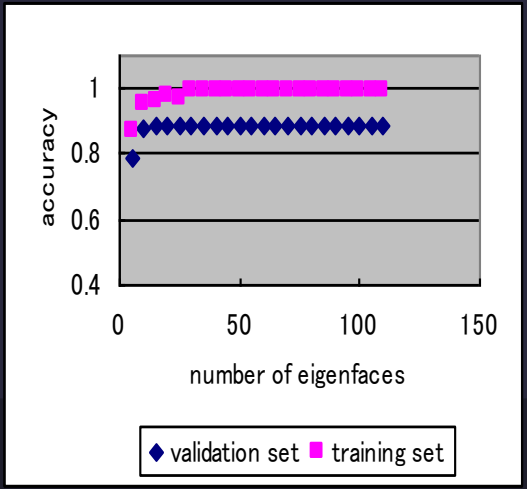
Classification Using Nearest Neighbor

- Save average coefficients for each person. Classify new face as the person with the closest average.
- Recognition accuracy increases with number of eigenfaces till 15.

  Later eigenfaces do not help much with recognition.

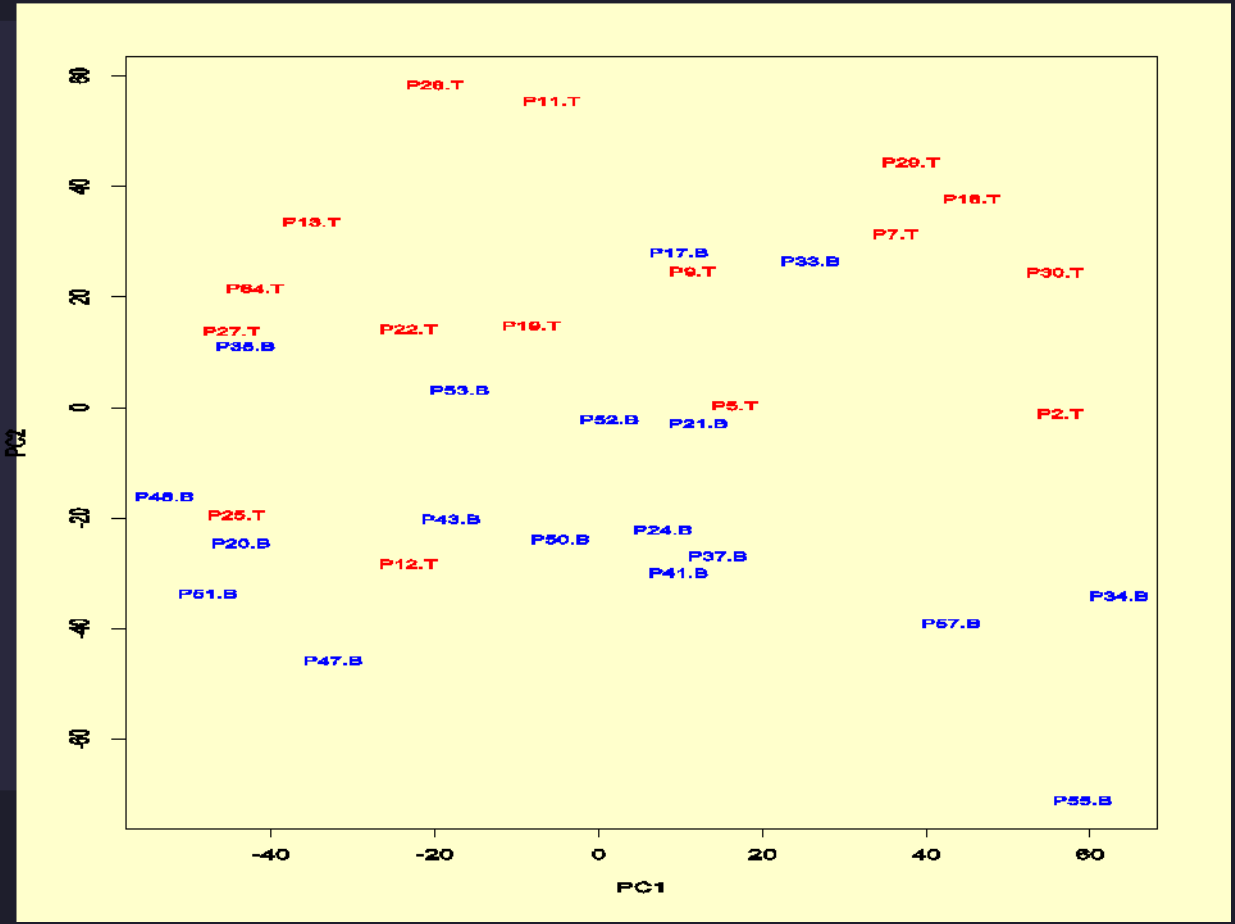
Best recognition rates

Training set	99%
Test set	89%



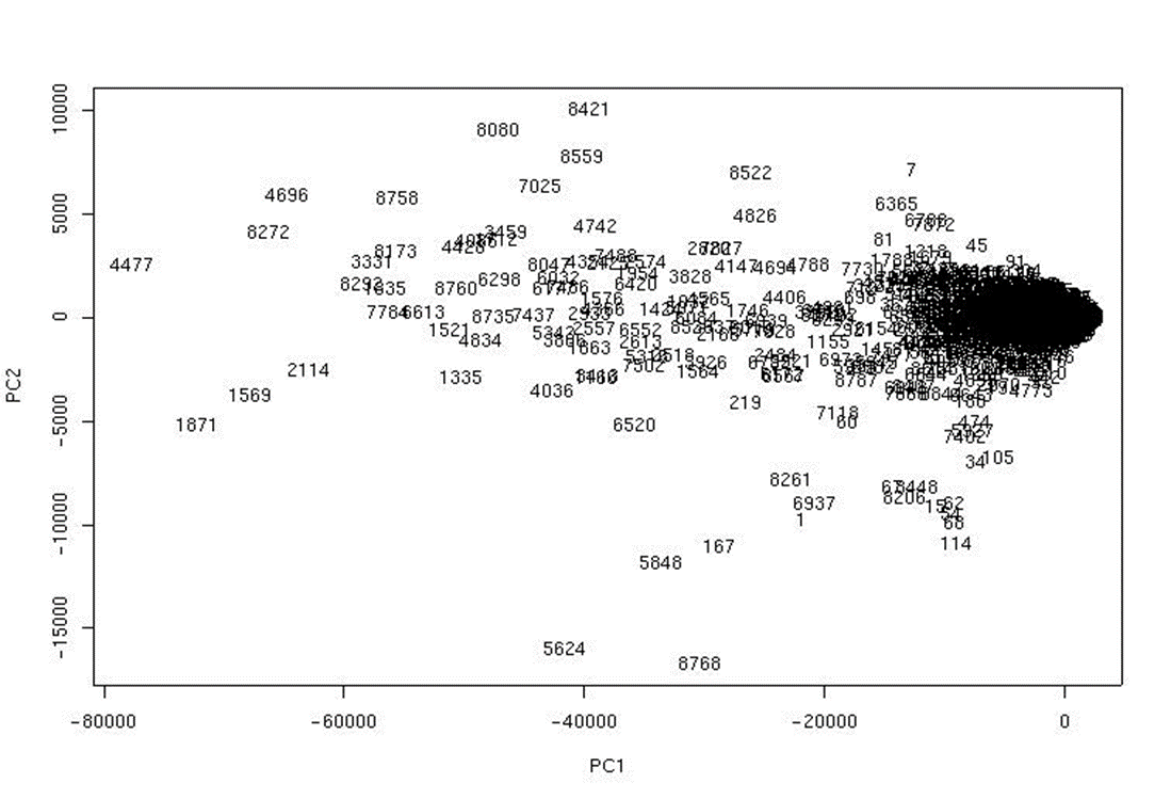
PCA of Genes (Leukemia data, precursor B and T cells)

34 patients, dimension of 8973 genes reduced to 2

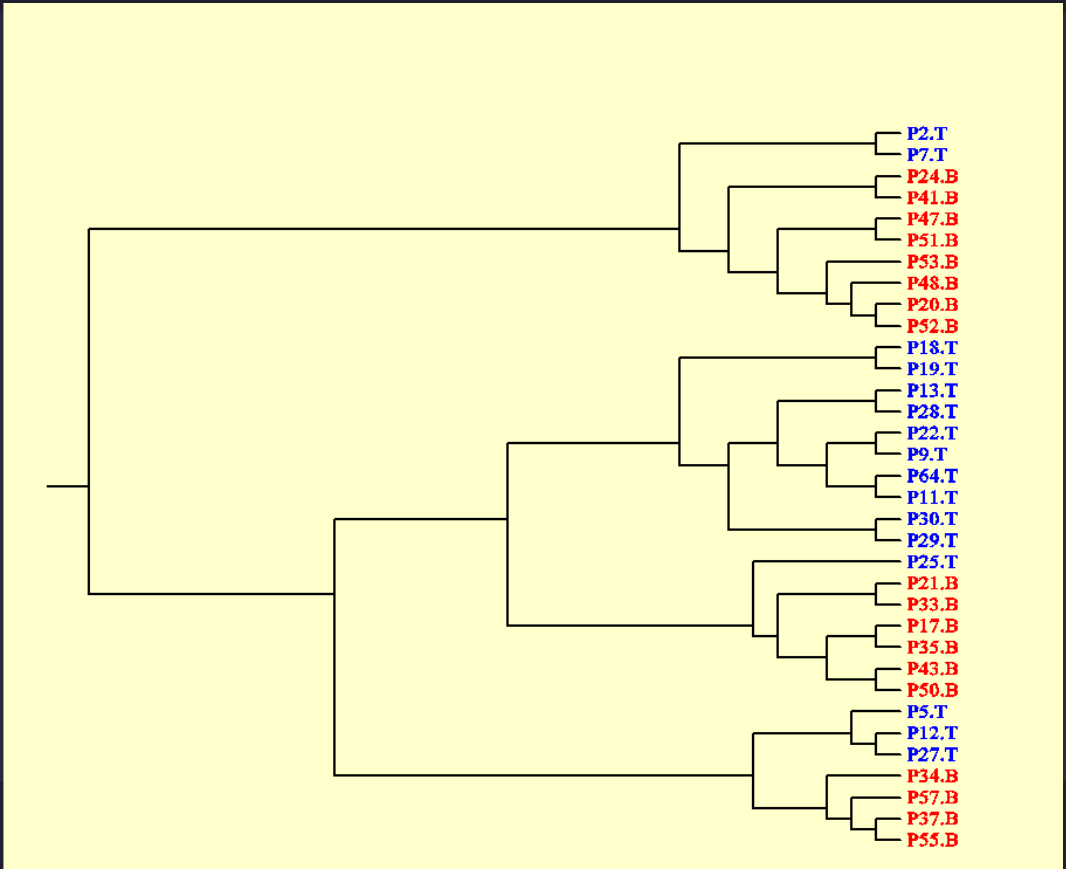


PCA of genes (Leukemia data)

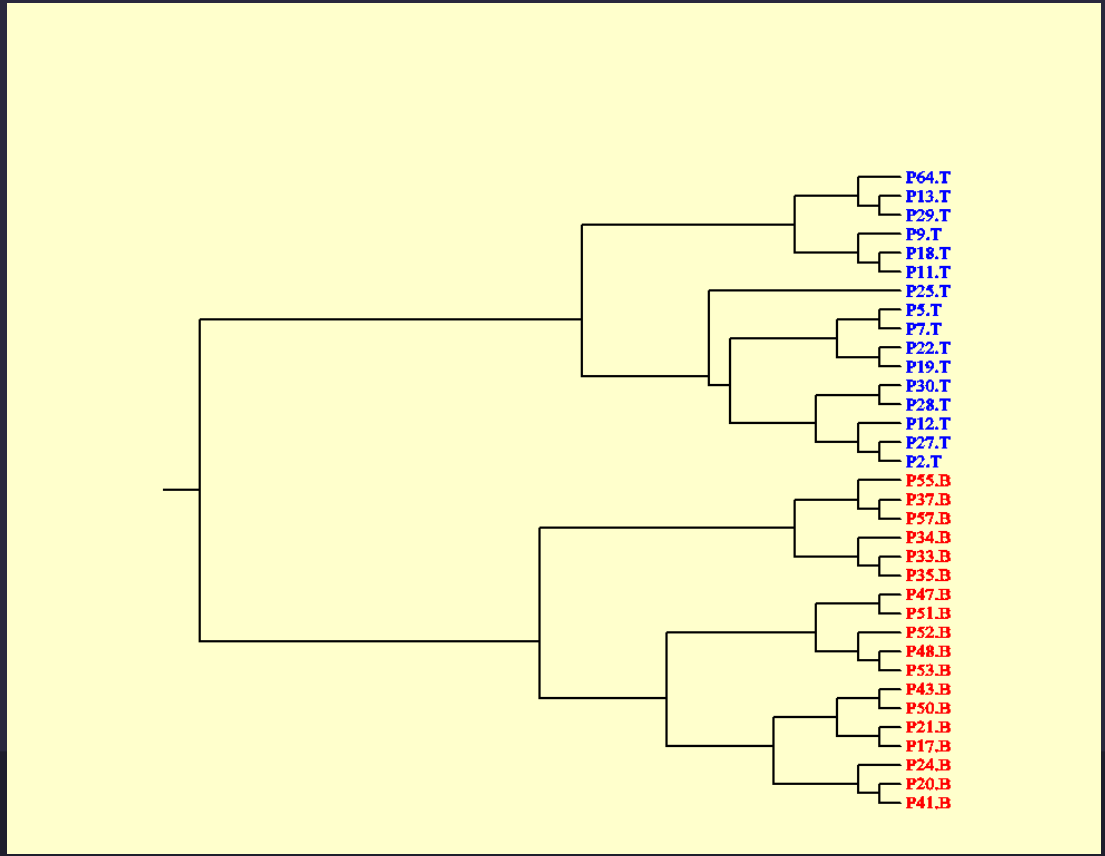
Plot of 8973 genes, dimension of 34 patients reduced to 2



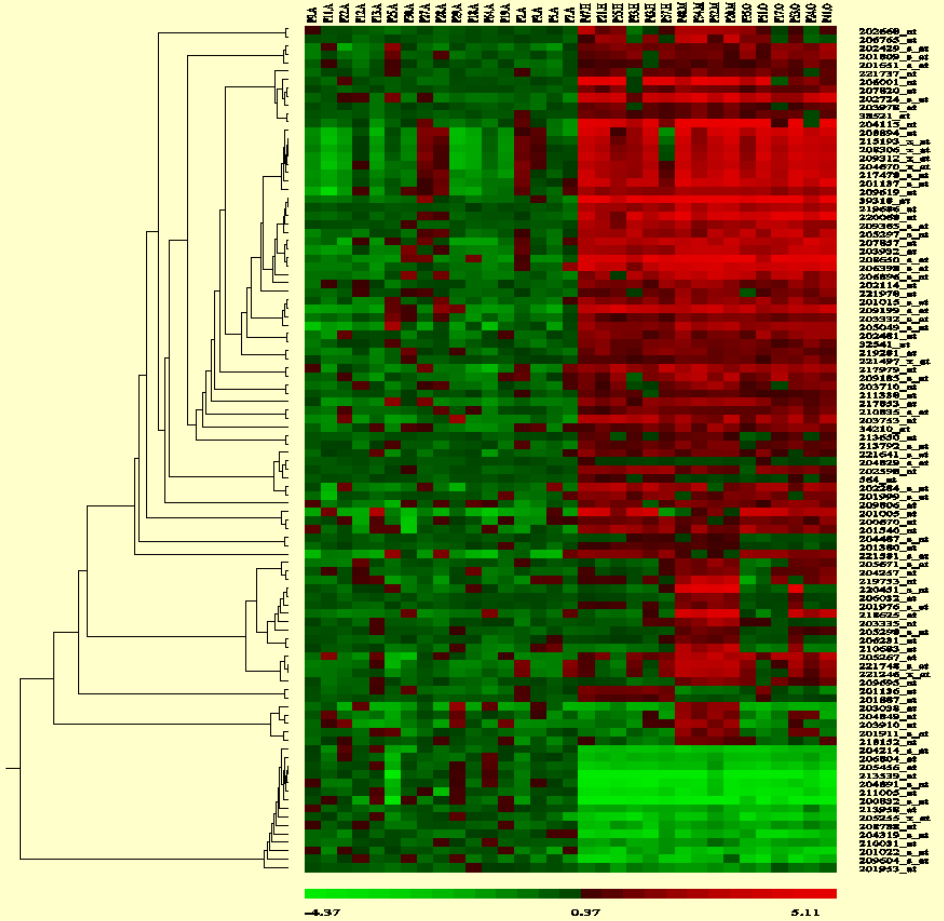
Leukemia data - clustering of patients on original gene dimensions



Leukemia data - clustering of patients on top 100 eigen-genes



Leukemia data - clustering of genes



ICA Solution and Applicability

- ICA is a statistical method, the goal of which is to decompose given multivariate data into a linear sum of statistically independent components.
- For example, given two-dimensional vector , $\boldsymbol{x} = [x_1 x_2]^T$, ICA aims at finding the following decomposition

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} s_1 + \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} s_2$$

$$\boldsymbol{x} = \boldsymbol{a}_1 s_1 + \boldsymbol{a}_2 s_2$$

where $\boldsymbol{a}_1$, $\boldsymbol{a}_2$ are basis vectors and s_1, s_2 are basis coefficients.

Constraint: Basis coefficients s_1 and s_2 are statistically independent.

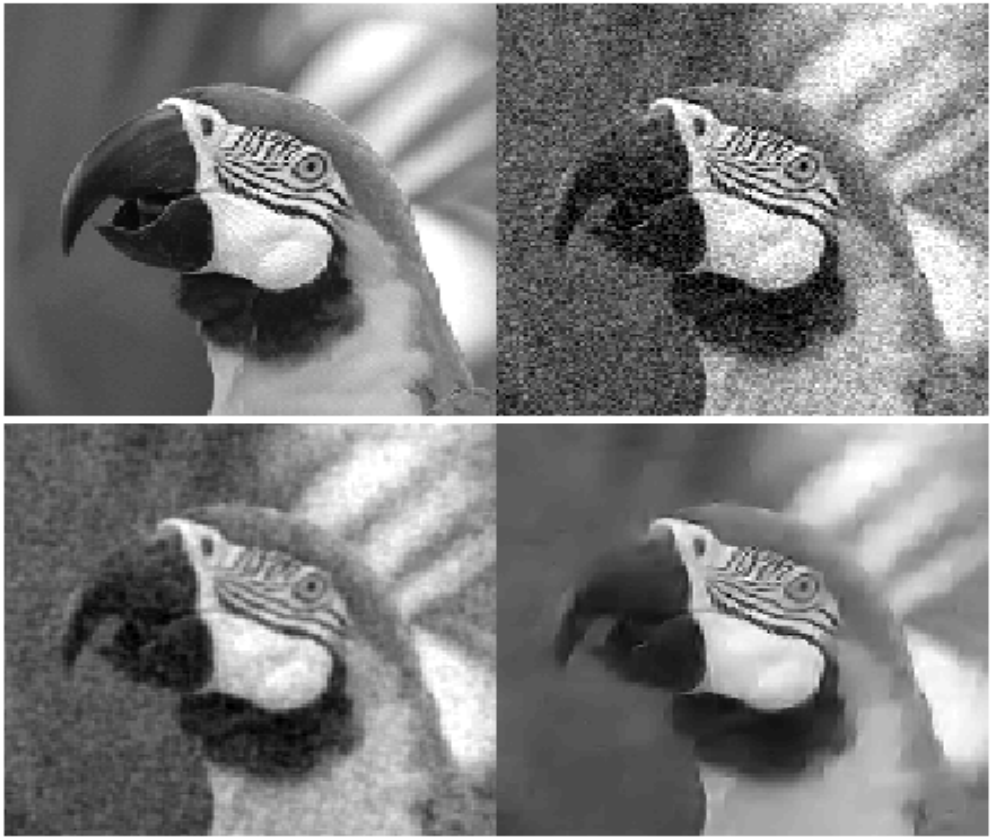
Application domains of ICA

- Blind source separation
- Image denoising
- Medical signal processing – fMRI, ECG, EEG
- Modelling of the hippocampus and visual cortex
- Feature extraction, face recognition
- Compression, redundancy reduction
- Watermarking
- Clustering
- Time series analysis (stock market, microarray data)
- Topic extraction
- Econometrics: Finding hidden factors in financial data

Image denoising

Original image

Noisy image



Wiener filtering

ICA filtering

Approaches to ICA

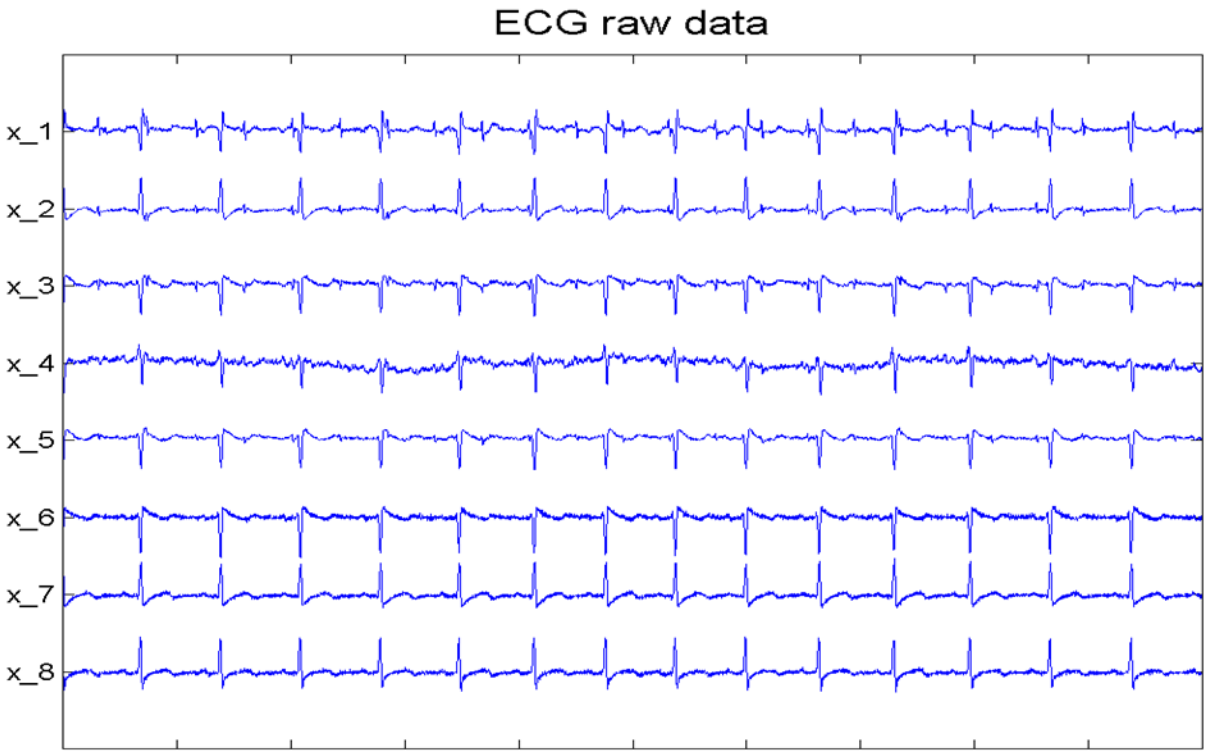
Unsupervised Learning

- Factorial coding (Minimum entropy coding, Redundancy reduction)
- Maximum likelihood learning
- Nonlinear information maximization (entropy maximization)
- Negentropy maximization
- Bayesian learning

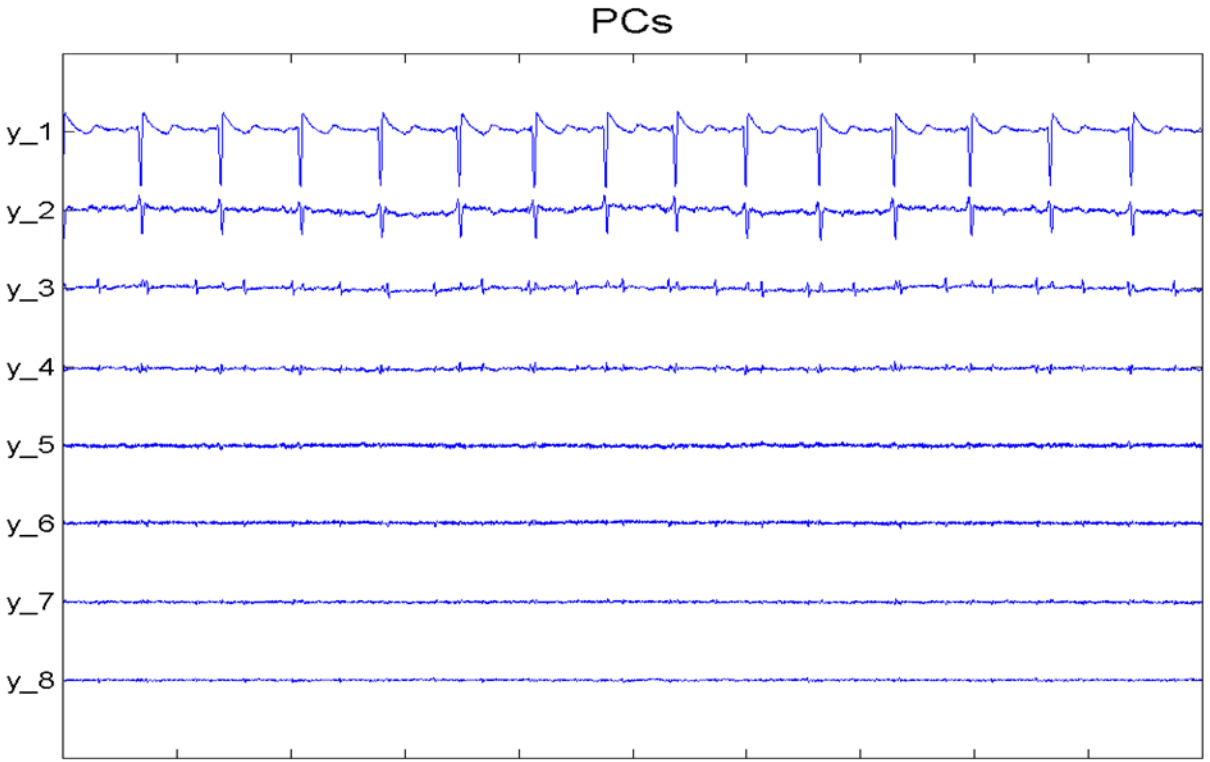
Statistical Signal Processing

- Higher-order moments or cumulants
- Joint approximate diagonalization
- Maximum likelihood estimation

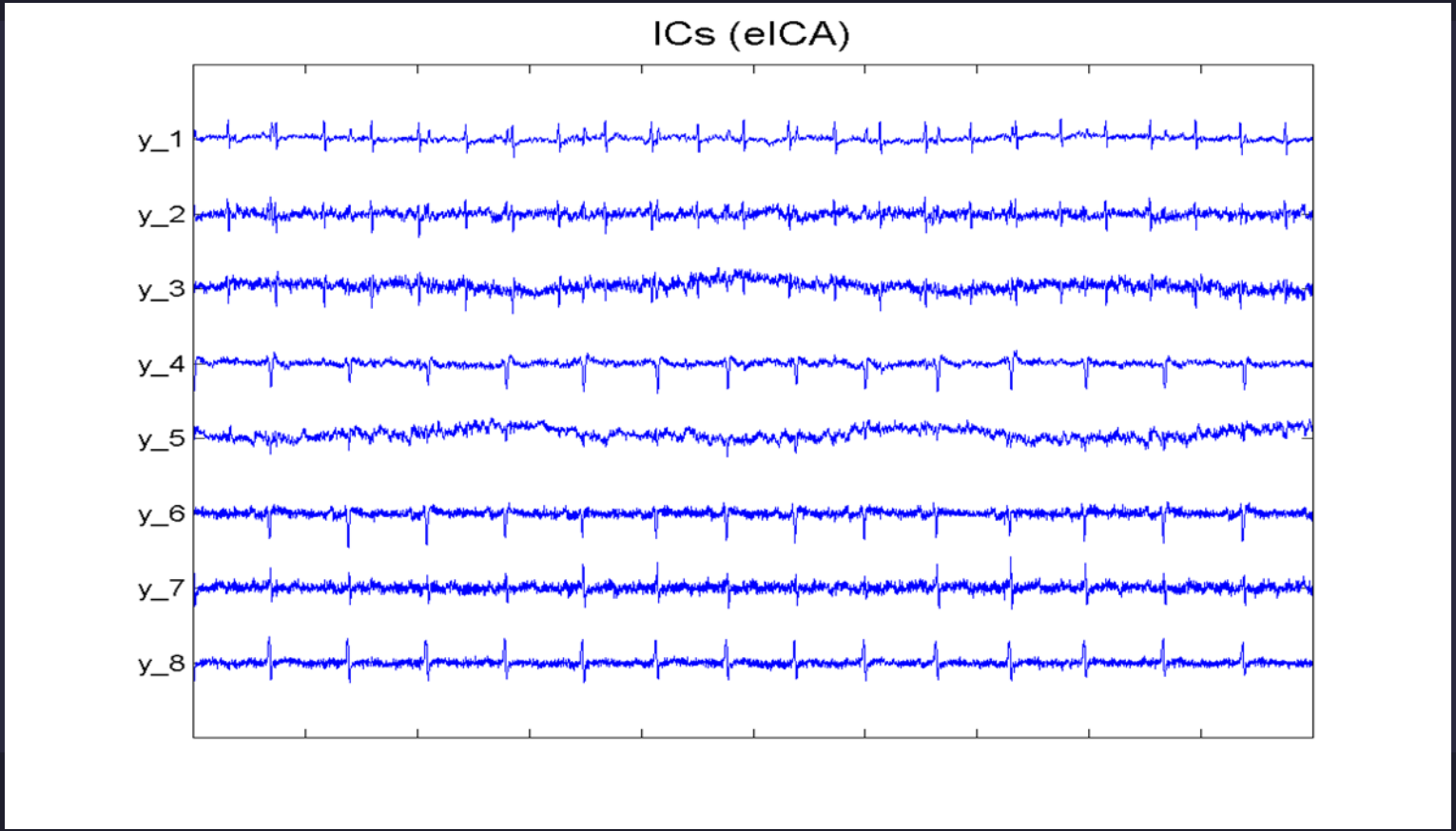
Feature Extraction in ECG data (Raw Data)



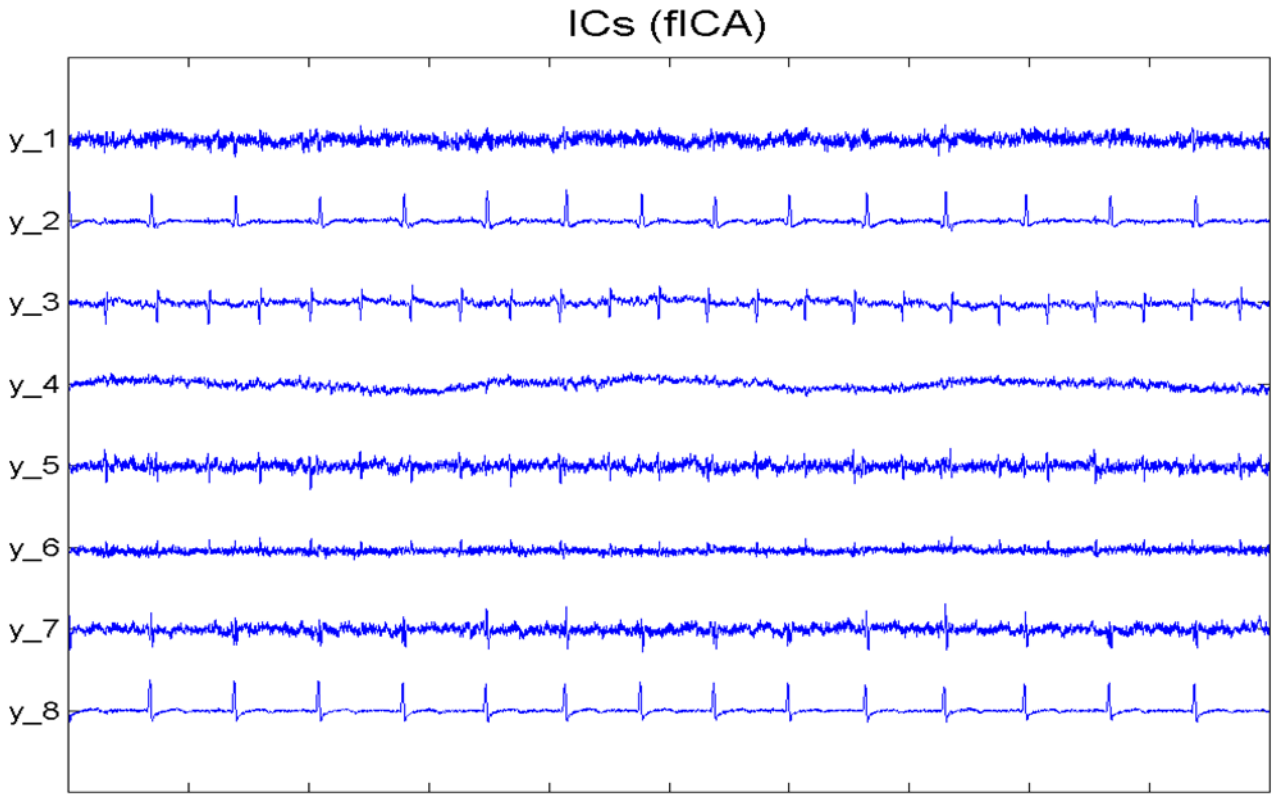
Feature Extraction in ECG data (PCA)



Feature Extraction in ECG data (Extended ICA)



Feature Extraction in ECG data (flexible ICA)



PCA vs ICA

Linear Transform

- Compression
- Classification

PCA

- Focus on uncorrelated and Gaussian components
- Second-order statistics
- Orthogonal transformation

ICA

- Focus on independent and non-Gaussian components
- Higher-order statistics
- Non-orthogonal transformation

Summary for Dimensionality reduction techniques

- PCA (Principal Component Analysis):
  Find projection that maximize the variance
- ICA (Independent Component Analysis):
  Very similar to PCA except that it assumes non-Guassian features
- Multidimensional Scaling:
  Find projection that best preserves inter-point distances
- LDA(Linear Discriminant Analysis):
  Maximizing the component axes for class-separation
- … etc

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<Thank You!>

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